

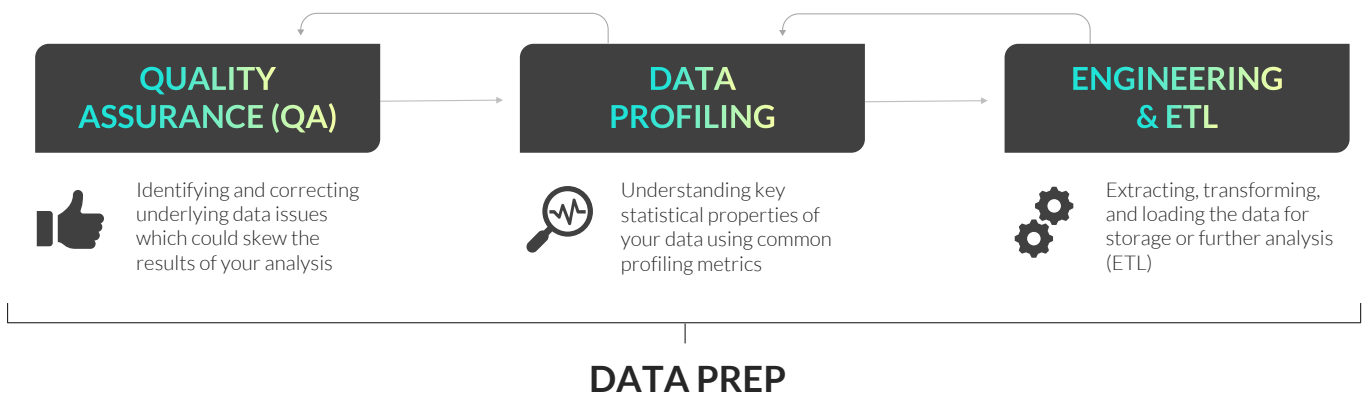
DATA PREP & ANALYSIS

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DATA PREP

GARBAGE IN, GARBAGE OUT

Data prep is about creating clean, quality data for analysis, and typically involves a combination of quality assurance (QA), basic profiling, and data engineering



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CASE STUDY: QUALITY ASSURANCE



THE SITUATION

You've been hired as a Data Analyst for **Maven Talent Hub**, a company that sends email newsletters to subscribers looking for job insights and opportunities.



THE ASSIGNMENT

You are interested in analyzing customer-level data to help the company improve audience targeting and segmentation, but the data is a mess.

Your task is to **QA the raw customer records** to identify any data quality issues that could potentially skew your analysis.



THE OBJECTIVES

1. Review a sample of user records exported from the company's CRM platform
2. Identify any potential data quality concerns
3. Suggest a potential solution for each issue you encounter

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QUALITY ASSURANCE (QA)

QUALITY ASSURANCE (QA)

DATA PROFILING

ENGINEERING & ETL

First Name	Last Name	City	State	Postal Code	Telephone	HH Income	Birthdate
Jennifer	Mcgrath	Fieldon	IL	62031	6,183,763,064	\$32,000	7/27/99
Susan	Rodriguez	Maplewood	NJ	7040	9,082,641,670	\$45,000	
Wayne	Nielson	Dahlgren	VA	22448	5,406,447,658	\$75,000	
John	Depaul	Baton Rouge	LA	70815	2,252,744,802	\$60,000	8/27/67
Joseph	Martinez	Portland	OR	97205	5,034,022,075	\$120,000	12/10/60
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Veronica	Comerford	Tuskegee	AL	36083	3,347,257,343	\$50,000	
Beverly	Nixon	Long Island	NY	11101	3,476,260,700	\$45,000,000	3/5/62
Ivan	Layton	Albany	New York	12207	5,184,317,602	\$105,000	
Debra	Ponce	Dallas	TX	75247	2,544,883,212	\$75,000	
Meg	Bryner	Providence	RI	2905	4,012,222,097	\$90,000	
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James	Weiss	Houston	TX	77060	8,322,324,190	\$135,000	
Addie	Stevenson	Jacksonville			9,042,482,500	\$80,000	
Laura	Hailey	Ephraim	Utah	84627	4,352,831,109	\$62,000	1/13/94
Daniel	Long						
Queen	Watson	Los Angeles	CA	90017	7,604,889,781	\$110,000	5/19/68

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! ISSUE: Incorrect data types (text formatted as values)

SOLUTION:

- Format or re-encode fields to ensure accurate and consistent data types (numeric, string, date, etc.)



PRO TIP: Fields containing dates, zip codes or phone numbers often cause problems!

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! ISSUE: Duplicate records

SOLUTION:

- Delete redundant records to avoid inflation or bias, unless duplicates serve a specific, valid purpose (i.e. up-sampling)

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! ISSUE: Inconsistent categorical values (*state name vs. abbreviation*)

SOLUTION:

- Determine a standard and find & replace inconsistent values
- Preserve the original data and add a new, standardized version for analysis



PRO TIP: Use **distinct count** to identify inconsistencies (*great for catching leading or trailing spaces!*)

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ENGINEERING & ETL

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! ISSUE: Empty values

SOLUTION:

- KEEP** if you are certain that the empty values are accurate and meaningful
- IMPUTE** (*substitute*) if you can accurately populate or estimate the data
- REMOVE** if you have a large volume of data and can't impute the empty values



PRO TIP: Never replace empty or NULL values with **0** unless you know it won't skew the data

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ENGINEERING & ETL

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⚠ **ISSUE:** Outlier value

SOLUTION:

- **KEEP** the outlier if it's valid and you want to capture its impact on the data
- **IMPUTE** (*substitute*) outliers with caps or replacement values
- **REMOVE** the outlier if it's an impossible value or significantly skews the analysis
- **TRANSFORM** values to reduce the effect of the outlier

In this case we see an average household income of **\$2,400,000** when we include the outlier. Does that feel accurate to you?

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DATA PROFILING METRICS

QUALITY ASSURANCE (QA)

DATA PROFILING

ENGINEERING & ETL

Count = 24
(Total # of Product SKUs)

Distinct Count = 8
(# of Unique Categories)

Conditional Count
(# Records by Category)

Snacks & Candy = 3
Frozen Foods = 3
Beverages = 3
Fruits & Vegetables = 3
Breakfast & Cereal = 3
Canned Goods = 3
Miscellaneous = 3
Dairy & Cheese = 3

SKU	Product Name	Category	Retail Price	Unit Cost	Margin
2827	Club Chocolate Bar	Snacks & Candy	\$1.25	\$0.38	\$0.88
4061	Big Time Cinnamon Waffles	Frozen Foods	\$3.75	\$2.57	\$1.19
4309	McGreevy's Apple Drink	Beverages	\$2.49	\$1.86	\$0.64
4637	High Top Oranges	Fruits & Vegetables	\$1.25	\$0.79	\$0.47
4967	Jeffers Oatmeal	Breakfast & Cereal	\$2.25	\$1.45	\$0.80
1285	Orchard Direct Frozen Fruits	Frozen Foods	\$2.99	\$1.66	\$1.33
4984	Toucan Canned Mixed Fruit	Canned Goods	\$1.49	\$0.40	\$1.09
4524	Best Choice Potato Chips	Snacks & Candy	\$1.99	\$1.77	\$0.22
5026	Club 1% Milk	Beverages	\$2.49	\$1.33	\$1.16
2874	Farmer's Choice Potatoes	Fruits & Vegetables	\$1.00	\$0.66	\$0.34
4554	Blue Label Black Beans	Canned Goods	\$1.25	\$1.15	\$0.10
4307	Acme Chip Clip	Miscellaneous	\$1.25	\$0.99	\$0.26
2472	Super Strawberry Preserves	Canned Goods	\$2.49	\$1.18	\$1.31
1893	Big Time Frozen Cauliflower	Frozen Foods	\$2.49	\$2.45	\$0.05
1715	Club Havarti Cheese	Dairy & Cheese	\$2.99	\$2.12	\$0.87
4476	Best Choice Peach Yogurt	Dairy & Cheese	\$1.75	\$1.25	\$0.51
4797	Lemon Dish Soap	Miscellaneous	\$3.99	\$2.97	\$1.02
2706	Honey Pot Apples	Fruits & Vegetables	\$1.00	\$0.54	\$0.47
2357	Club String Cheese	Dairy & Cheese	\$1.99	\$1.15	\$0.84
4599	Sunrise Corn Splats	Breakfast & Cereal	\$1.49	\$1.04	\$0.45
1931	Blue Ribbon Tonic Water	Beverages	\$1.00	\$0.69	\$0.32
3568	Best Choice Fruit Roll	Snacks & Candy	\$2.75	\$1.50	\$1.26
4363	Top Choice Wheat Bites	Breakfast & Cereal	\$2.75	\$2.69	\$0.06
4603	Denny Copper Pot Scrubber	Miscellaneous	\$3.75	\$3.12	\$0.63

Range = \$1.00 - \$3.99
(Max & Min Retail Price)

Mean = \$2.16
(Average Retail Price)

Median = \$2.12
(“Middle” Retail Price)

Mode = \$1.25
(Most Frequently Observed Price)

Percentile = 91%
(% of Records where **Margin** < \$1.26)

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DATA PROFILING VISUALS

QUALITY ASSURANCE (QA)

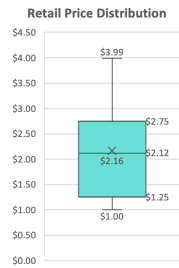
DATA PROFILING

ENGINEERING & ETL



Histogram

Shows the frequency distribution of a data set, and can be used to identify outliers or compare profiling metrics like variance or skew



Box Plot

Shows the minimum, maximum, median, mean, and 25th/75th quartile values, and can be used to plot outliers or compare distributions across categories



Scatter Plot

Shows the relationship between two numerical variables, and can be used to identify correlation or impute missing values

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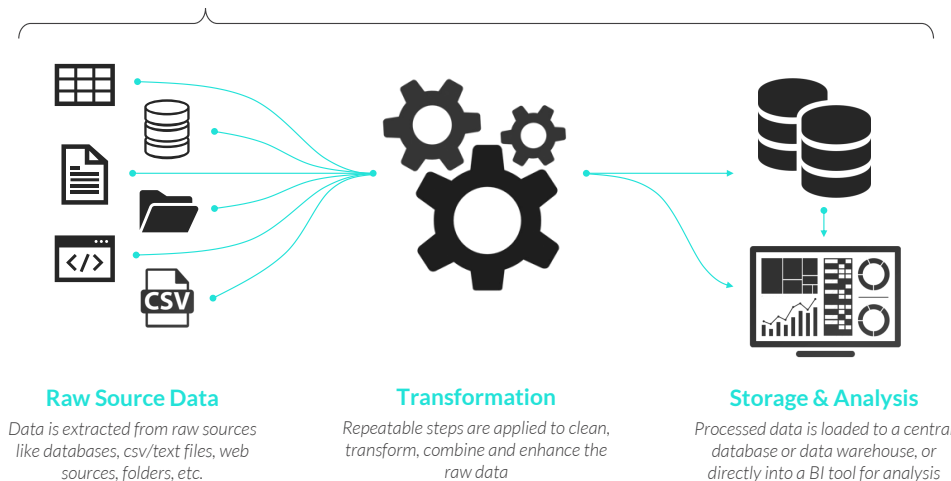
DATA ENGINEERING & ETL

QUALITY ASSURANCE (QA)

DATA PROFILING

ENGINEERING & ETL

ETL is a term used to describe the process of **extracting**, **transforming** and **loading** raw data for centralized storage or analysis



Common ETL transformations:

- ✓ Feature engineering
- ✓ Filtering
- ✓ Cleaning
- ✓ Merging, appending & joining
- ✓ Transposing
- ✓ Grouping & aggregating
- ✓ Normalizing

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DATA ANALYSIS



EXPLORATORY (EDA)

"How does performance compare across different segments or categories?"

"Are there any interesting correlations or relationships between variables?"

"How is the data distributed, and are there any outliers or missing values?"

"How are key metrics changing over time? Are there predictable seasonal trends?"



EXPLANATORY

"Why did Conversion Rate drop by 25% month-over-month?"

"Which products have seen the largest increases in gross revenue this year?"

"Which customer segments deliver the most lifetime value (LTV)?"

"Based on geographical trends, where should we open our next store location?"

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DATA-DRIVEN INSIGHTS

TURN INFORMATION INTO **INSIGHT**

A strong, data-driven insight should communicate **what** happened, **why** it happened, and **how** you recommend taking action (*this one is key!*)



Tips for writing effective insights:

- ✓ Tell a clear and logical story
- ✓ Support your findings with hard data and facts
- ✓ Provide meaningful context behind key data points
- ✓ Include actionable recommendations and next steps
- ✓ Remember that it's all about **driving business outcomes**

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DATA-DRIVEN INSIGHTS



“

*This week we sold **80** pairs of socks*

”



✓ Explains what happened (sort of)

- Doesn't provide any meaningful context
- Doesn't explain what drove the sales
- Offers no recommendation or plan of action

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DATA-DRIVEN INSIGHTS



“

*This week we sold **80** pairs of socks
(+14%), while spend increased **25%***

”



✓ Explains what happened

✓ Provides context in terms of weekly change and comparison against spend

- Doesn't explain what drove the sales or the spend increase
- Offers no recommendation or plan of action

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DATA-DRIVEN INSIGHTS



“

This week we sold **80** pairs of socks (+14%), while spend increased **25%**

- New ads drove the incremental sales, at **\$7/sale** (vs. \$4 benchmark)

”



- ✓ Explains what happened
- ✓ Provides context in terms of weekly change and comparison against spend
- ✓ Explains that new ads drove the incremental sales, but at a high cost vs. benchmark
- Offers no recommendation or plan of action

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DATA-DRIVEN INSIGHTS



“

This week we sold **80** pairs of socks (+14%), while spend increased **25%**

- New ads drove the incremental sales, at **\$7/sale** (vs. \$4 benchmark)
- **Recommendation:** Pause new ads and allocate budget to more efficient channels

”



- ✓ Explains what happened
- ✓ Provides context in terms of weekly change and comparison against spend
- ✓ Explains that new ads drove the incremental sales, but at a high cost vs. benchmarks
- ✓ Offers a clear, actionable, data-driven recommendation

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KEY TAKEAWAYS



QA comes first. **Every. Single. Time.**

- *Underlying data issues can significantly skew the results of your analysis, so take QA seriously!*



Before you analyze your data, **understand** it first

- *Use profiling metrics and tools like box plots or histograms to explore statistical properties of the data*



Create a repeatable plan to **extract, transform & load** data for analysis

- *Many self-service BI tools include powerful, native ETL capabilities (like Power Query and Tableau Prep)*



Data analysis can be categorized as either **exploratory** or **explanatory**

- *Exploratory data analysis (EDA) is about discovering insights; explanatory analysis is about telling a story*



Insights should always lead to **clear, actionable recommendations**

- *If your insights don't drive measurable improvements to the business, what purpose do they serve?*